

Phase F: Final Security Assessment & ISO/IEC 27001 Alignment

AWS CLI Security and IAM Assessment

Phase Objective

Compile all findings from the previous phases into a professional cloud security report, identify key AWS cloud security risks, and align the findings with relevant ISO/IEC 27001 security and access-control principles.

Actions Taken

In Phase F, I consolidated the results and evidence collected throughout Phases A–E into a final cloud security assessment covering AWS IAM identity management, AWS CLI authentication, user permissions, Boto3-based access, MFA, and temporary AWS credentials.

Phase A demonstrated AWS CLI authentication and identity verification using AWS STS. Phase B focused on configuring separate IAM users and AWS CLI profiles with different permission levels.

Phase C validated permissions by testing EC2 and S3 operations with dev-user and ops-user. The restricted user was denied unauthorized actions while the administrative user was permitted to perform the tested operations.

Phase D focused on MFA and temporary session credentials. The temporary credentials were configured under the ops-temp profile and validated using AWS STS. The expiration information demonstrated that temporary credentials have a limited lifetime.

I reviewed risks including credential leakage, excessive permissions, unauthorized access, privilege escalation, and long-term credential exposure.

I reviewed security controls including least-privilege IAM policies, MFA, temporary credentials, separate AWS CLI profiles, and secure credential management.

The final report was structured to document methodology, technical findings, evidence, threat relationships, security controls, and recommendations, with consideration of relevant ISO/IEC 27001 principles.

Key Observations

- Least privilege limited the actions available to dev-user and reduced the potential impact of unauthorized access.
- Administrative access provided ops-user with broader permissions, demonstrating why privileged accounts require stronger protection.
- MFA adds an additional authentication factor and reduces the risk associated with compromised credentials.
- Temporary AWS credentials reduce the exposure window because they automatically expire.
- Credential leakage and excessive permissions are significant risks in cloud environments.

- IAM access should be regularly reviewed to ensure users have only the permissions required for their responsibilities.
- AWS IAM and credential-management practices can support broader ISO/IEC 27001-aligned access-control and authentication objectives.

Threat and Risk Assessment

Risk	Potential Impact	Security Control
Credential leakage	Unauthorized access to AWS resources	Secure credential storage, credential rotation, MFA
Excessive IAM permissions	Increased impact of compromised accounts	Least privilege and regular permission reviews
Compromised user credentials	Attacker gains access using a valid account	MFA and temporary credentials
Privilege escalation	Unauthorized increase in access privileges	Restrictive IAM policies and monitoring
Long-term static credentials	Extended unauthorized access if keys are stolen	Temporary credentials and regular key rotation

ISO/IEC 27001 Alignment

Security Area	Application to This Task
Access Control	IAM users were assigned different permissions according to their required access levels.
Identity Management	AWS CLI profiles and STS were used to identify and validate authenticated AWS identities.
Authentication Information	Access keys and credentials were treated as sensitive information requiring secure management.
Secure Authentication	MFA was implemented to provide an additional authentication factor.
Least Privilege	The restricted dev-user was prevented from performing unauthorized EC2, S3, and IAM operations.
Temporary Access	STS temporary credentials were used to reduce reliance on long-term credentials.
Access Review	Permission testing demonstrated the importance of validating whether users can perform only their intended actions.

Conclusion

Phase F consolidated the technical results from the AWS cloud security exercises into an overall security assessment. The practical testing demonstrated that IAM permissions directly control access to AWS resources and that least-privilege configurations can reduce unauthorized actions. The implementation of MFA and temporary credentials further demonstrated effective methods for reducing the risks associated with compromised or exposed credentials.

Overall, the task showed that secure AWS identity and access management requires more than simply authenticating users. Effective cloud security also depends on enforcing appropriate permissions, protecting credentials, using strong authentication controls, limiting credential lifetimes, and regularly reviewing access. These practices contribute to a stronger AWS security posture and align with broader ISO/IEC 27001 principles for access control and information security management.

Evidence Produced

- Final_AWS_CLI_Security_Report.pdf — Final report consolidating findings, methodology, evidence, risk assessment, and ISO/IEC 27001 alignment.
- Phase A evidence — AWS CLI identity verification and IAM credential analysis.
- Phase B evidence — IAM user creation and AWS CLI profile configuration.
- Phase C evidence — EC2 and S3 permission testing, including permitted and denied access results and Boto3 testing.
- Phase D evidence — MFA and temporary credential evidence, including token expiration output.
- Phase E evidence — Completion certificates for the required hosted AWS CLI training labs.